

DD_FA-02 - Network Field Audit - Developer Implementation Guide

Version: v1.0 Bundle: 10_other Companion DD: DD_FA-02-Network_Field_Audit-v1.0

1. Goal

Implement network field audits without duplicating RLM, WO, or provisioning ownership.

FA-02 stores audit tasks, expected snapshots, observations, discrepancies, and routing. RLM-CFG-01 remains the source of truth for HomePass/topology, WO-01 handles dispatch, and PROV-INT provides network read/check data.

2. Runtime Flow

```
sequenceDiagram
    participant B0 as Backoffice
    participant FA as FA-02
    participant RLM as RLM-CFG-01
    participant PROV as PROV-INT
    participant WO as WO-01
    participant MOB as Mobile App
    participant APR as EM-CFG-04

    B0->>FA: POST /api/network-field-audits/tasks
    FA->>RLM: read expected HomePass/topology
    FA->>PROV: optional read signal/port state
    FA->>FA: store expected snapshot
    FA->>WO: optional create FIELD_NETWORK_AUDIT WO
    MOB->>FA: submit GPS/photos/measurements
    MOB->>FA: submit task
    FA->>FA: compare expected vs observed
    FA->>FA: run Drools severity/routing
    FA->>APR: approval if sellability/topology correction needs it
    FA->>RLM: request approved correction
```

3. Step 1 - Migrations

Create:

- network_field_audit_campaign
- network_field_audit_task
- network_field_audit_expected_state
- network_field_audit_observation
- network_field_audit_discrepancy

Add unique (operator_code, offline_client_ref) for observations to support offline mobile retry.

4. Step 2 - Create Task

Endpoint:

```
POST /api/network-field-audits/tasks
Idempotency-Key: nfac-HP-GPON-0440-202608
Content-Type: application/json
```

Request:

```
{
  "operatorCode": "WIK",
  "campaignId": "nfac-001",
  "taskType": "HOMEPASS",
  "networkRefType": "HOMEPASS",
  "networkRefId": "HP-GPON-0440",
```

```

    "homepassId": "HP-GPON-0440",
    "assignedToUserId": "usr-tech-020",
    "createWorkOrder": true
  }

```

FA-02 behavior:

1. Validate permission through EM-CFG-03.
2. Apply idempotency by key and request hash.
3. Insert task in CREATED.
4. Call RLM-CFG-01 to fetch expected network state.
5. Optionally call PROV-INT read APIs for signal/port state.
6. Insert network_field_audit_expected_state rows.
7. If createWorkOrder=true, create WO.
8. Emit NetworkFieldAuditTaskCreated.

5. Step 3 - Create WO

Call WO-01:

```

POST /api/work-orders
Idempotency-Key: nfat-001-wo
Content-Type: application/json

{
  "operatorCode": "WIK",
  "kind": "FIELD_NETWORK_AUDIT",
  "originatingContextType": "NETWORK_FIELD_AUDIT_TASK",
  "originatingContextId": "nfat-001",
  "homepassId": "HP-GPON-0440",
  "metadata": {
    "networkRefType": "HOMEPASS",
    "networkRefId": "HP-GPON-0440",
    "requiredEvidence": ["gps", "photo", "signalMeasurement", "portCount"]
  }
}

```

Store wo_id and move task to ASSIGNED.

6. Step 4 - Mobile Observation

Endpoint:

```

POST /api/network-field-audits/tasks/nfat-001/observations
Idempotency-Key: mob-nfa-991
Content-Type: application/json

```

Request:

```

{
  "observationCode": "OPTICAL_POWER_TEST",
  "observedValue": {
    "rxPowerDbm": -19.4,
    "freePorts": 4
  },
  "conditionStatus": "OK",
  "photoFileIds": ["file-101"],
  "gpsLatitude": -1.2920,
  "gpsLongitude": 36.8218,
  "capturedAt": "2026-08-16T10:15:00Z",
  "offlineClientRef": "mob-nfa-991"
}

```

FA-02 behavior:

1. Verify task assignment.
2. Validate uploaded file ids through file storage metadata.
3. Store observation.

4. Return the existing observation on duplicate offline ref.

7. Step 5 - Submit And Compare

Endpoint:

```
POST /api/network-field-audits/tasks/nfat-001/submit
```

Comparison:

1. Load expected snapshots.
2. Load observations.
3. Detect MISSING_NODE, WRONG_GPS, LOW_SIGNAL, PORT_MISMATCH, NOT_SERVICEABLE, or UNDOCUMENTED_PLANT.
4. Create discrepancies.
5. If none, close task.
6. If any, run routing.

8. Step 6 - Drools

Fact:

```
public record NetworkFieldAuditFact(  
    String operatorCode,  
    String discrepancyType,  
    String technologyFamily,  
    boolean homepassSellable,  
    int usablePortCount,  
    double rxPowerDbm,  
    String proposedSellability  
) {}
```

Rules:

```
package rules.field_audit.network.severity  
  
rule "No usable port on sellable HomePass"  
when  
    $f : NetworkFieldAuditFact(discrepancyType == "PORT_MISMATCH",  
                               homepassSellable == true,  
                               usablePortCount == 0)  
    $d : NetworkFieldAuditDecision()  
then  
    $d.setSeverity("HIGH");  
    $d.route("REQUEST_RLM_CORRECTION");  
end  
  
package rules.field_audit.network.approval  
  
rule "Sellability downgrade requires approval"  
when  
    $f : NetworkFieldAuditFact(proposedSellability == "NOT_SELLABLE")  
    $d : NetworkFieldAuditDecision()  
then  
    $d.requireApproval("NETWORK_FIELD_AUDIT_SELLABILITY_CHANGE");  
end
```

9. Step 7 - Route

If route is REQUEST_RLM_CORRECTION, call EM-CFG-04 first when required, then call RLM:

```
POST /api/homepass-corrections  
Idempotency-Key: nfad-001-rlm-correction  
Content-Type: application/json  
  
{  
    "operatorCode": "WIK",  
    "originRefType": "NETWORK_FIELD_AUDIT_DISCREPANCY",  
    "originRefId": "nfad-001",  
    "homepassId": "HP-GPON-0440",  
    "correctionType": "SELLABILITY_CHANGE",  
}
```

```
    "proposedValue": {  
      "sellability": "NOT_SELLABLE",  
      "reason": "No usable port found during audit"  
    }  
  }
```

For customer-impacting issues, create a TCK ticket instead of directly contacting the customer from FA-02.

10. Tests

1. Create task freezes RLM snapshot.
2. Duplicate task request returns same task.
3. WO creation carries correct origin context.
4. Observation idempotency works offline.
5. Port mismatch opens discrepancy.
6. Sellability correction requires EM-CFG-04 when configured.
7. RLM is called only through API, not local DB writes.